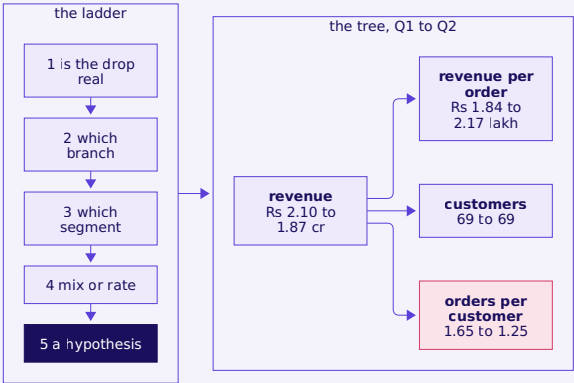


The investigation ladder

Kalpa Retail, Week 1 Tuesday. A drop is investigated in a fixed order, and each rung is a question that has to be answered before the next one is worth asking. The tree beside the ladder carries the closed quarters, Q1 against Q2, on booked orders as exported.

1 The ladder beside the tree



Revenue fell 11.0 percent between two closed quarters. Customers held at 69, orders per customer fell 24.6 percent, and revenue per order rose 18.0 percent, so the ratios multiply back: 1.000 times 0.754 times 1.180 is 0.890. In behaviour the fall sits in Retail-Plus, where the same 22 members placed 26 orders against 51.

Crux. A rate without its denominator is a rumour.

2 Windows, like with like

Window	Result	Verdict
Q1 against Q2 cut at 15 Sep	Down 25.9 percent	First and last dates show 13 weeks against 11.
Closed quarters, 13 weeks each	Down 11.0 percent	It is fair once Q2 has closed.
Per week, Q2 still open	Down 12.4 percent	A rate makes it comparable.

Crux. Confirm the drop on matched windows before you explain it.

3 The tree, as a bridge in rupees

Step	Rupees
Q1 revenue	Rs 2,10,00,000
Customers, 69 to 69	Rs 0
Orders per customer, 28 fewer orders	Minus Rs 51,57,895
Revenue per order, Rs 33,231 more on 86	Plus Rs 28,57,895
Q2 revenue	Rs 1,87,00,000

Crux. Decompose along the tree: customers, orders per customer, revenue per order.

4 Missing is unknown

```
order.get("discount", 0) # a decision that looks like no decision
```

Count the records that carry the field, average only where it is recorded, and bound the branch with the largest recorded value: Rs 150 on 86 orders is at most Rs 12,900 against a Rs 23,00,000 fall.

Crux. Missing means unknown until someone chooses a default and writes down why.

5 Roll-up with weights, and mix against rate

A plain average of four segments gives 1.94 and 1.82; total orders over total customers gives 1.65 and 1.25, which is the figure the roll-up must reproduce. Revenue per order rose Rs 33,231, and about 69 percent of it is mix, because small Retail-Plus orders fell from 44.7 to 30.2 percent of orders.

Crux. Roll a rate up with its weights; never average the averages.

6 Functions return

```
def tree_for(rows):  
    ...  
    return {"revenue": revenue, "orders": n, "customers": c}
```

A helper that prints for large changes returns *None*, the filter drops it, and four segments go in while two rows come out. Check with *None* in `changes.values()`.

Crux. A function returns its answer; count the groups in and the groups out.

7 Hypothesis with evidence

Test	What the file shows
Timing	Retail-Plus fell from July, before the late-August break the complaint dates.
Channel	Web, store and app all fell, so the app did not fall first and alone.
Evidence to ask for	The app's reorder logs by week, the release date, and the tier's change log.

Crux. Name the cause as a hypothesis, with the evidence that would settle it.

THE WORDS ON THIS SHEET

Matched windows Two periods of the same length compared, or both turned into a rate

Rate A numerator over a denominator in a stated window

Decomposition A change split along the tree into branches that multiply

Bridge A change in rupees moved one branch at a time from start to end

Default The value used when a field is absent, with its written reason

Function A named block that takes inputs and returns one answer